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Short-term capture of the Earth–Moon system

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ABSTRACT

In this paper, the short-term capture (STC) of an asteroid in the Earth–Moon system is proposed and investigated. First, the space condition of STC is analysed and five subsets of the feasible region are defined and discussed. Then, the time condition of STC is studied by parameter scanning in the Sun–Earth–Moon–asteroid restricted four-body problem. Numerical results indicate that there is a clear association between the distributions of the time probability of STC and the five subsets. Next, the influence of the Jacobi constant on STC is examined using the space and time probabilities of STC. Combining the space and time probabilities of STC, we propose a STC index to evaluate the probability of STC comprehensively. Finally, three potential STC asteroids are found and analysed.

Key words: methods: numerical – celestial mechanics – minor planets, asteroids: general – planets and satellites: general.

1 INTRODUCTION

Recently, the study of asteroids has become a hot topic in aerospace science and has attracted significant attention from the researchers. The investigation of asteroids can not only help us unravel the origin of the Solar system, but can also be used to establish early warning of asteroid impact (Tonry 2010; Farnocchia, Chesley & Micheli 2015; Cheng et al. 2016). Several asteroid retrieval missions are conducted or proposed, such as JAXA's Hayabusa, and NASA's Asteroid Redirect Robotic Mission (ARRM). Among numerous asteroids, the near Earth asteroids (NEAs) with perihelion less than 1.3 astronomical unit (au) are more interesting and important. Because the NEAs are close to the Earth, they are the most feasible asteroids for retrieval or capture missions. The research of Hartmann & Sokolov (1994) showed that a 2 km-sized metallic NEA may contain rich metals and materials worth more than 25 trillion dollars. The main objective of NASA's ARRM is to capture an entire 4–10 m NEA and redirect it into an orbit around the Moon where astronauts in the Orion spacecraft could explore it (Strange et al. 2013). However, the masses of the NEAs are usually quite large, for example, the capacity of the ARRM spacecraft is designed to return an asteroid with a mass of up to 1000 t (10^6 kg) to the Earth–Moon system (EMS), and in the research of Sánchez & Yárnoz (2016), the retrievable masses ranges between 277 and 2100 t. The large NEA masses result in huge fuel costs for capture missions, especially at the endgame of these missions when the NEAs need to brake and be captured in the EMS (Hasnain, Lamb & Ross 2012). Hence, low-energy capture orbits for NEAs are proposed and studied by many researchers.

Baoyin, Chen & Li (2010) proposed a method to search for low-energy NEAs that may be temporarily captured by the Earth. Their results indicated that some NEAs could be captured by the Earth after a small velocity increment (less than 1 km s^{-1}) while close to the Earth. A family of so-called Easily Retrievable Objects, which can be transported from accessible heliocentric orbits into the Earth's neighbourhood at affordable costs was defined and investigated by Yarnoz Garcia, Sanchez & McInnes (2013). The asteroid retrieval missions from the heliocentric orbits of the candidate asteroids to the libration point orbits (LPOs) in the Sun–Earth circular restricted three-body problem (CRTBP) were proposed and investigated by incorporating low-thrust propulsion into the invariant manifolds technique (Ceriotti & Sanchez 2016; Sánchez & Yárnoz 2016; Tang & Jiang 2016). Tan, McInnes & Ceriotti (2017) investigated the design problem of the asteroid retrieval mission to LPOs around L_2 in the Earth–Moon CRTBP. Bao et al. (2015) applied a lunar gravity assist (LGA) and an Earth gravity assist to capture NEAs into bounded orbits around the Earth. Verrier & McInnes (2014) studied the low-energy capture of asteroids on to Kolmogorov–Arnold–Moser tori via the chaos-assisted capture mechanism in the planar Hill problem of the Sun–Earth system.

In reality, the capture problem is not a recent issue, for example, the lunar capture problem of Earth-to-Moon transfers has been studied by numerous scholars. The most famous one is the gravitational capture or the weak stability boundary (WSB) theory first proposed by Belbruno (1987). The core idea of the WSB or gravitational capture is using the gravities in the *N*-body problem to achieve a temporary capture (TC) without manoeuvres based on propulsive systems. Although this capture is just temporary, a permanent capture will be accomplished with less fuel consumption if the spacecraft is braked during this TC. This theory has been successfully applied to many lunar missions, such as JAXA's Hiten

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rescue (Belbruno & Miller 1990), ESA's SMART-1 (Schoenmaekers, Horas & Pulido 2001), and NASA's GRAIL (Roncoli & Fujii 2010). Furthermore, it is also extended to some studies about interplanetary transfers, such as to Mercury (Jehn et al. 2004) and Mars (Topputo & Belbruno 2015). A series of successful applications of the WSB inspire us to extend it to the asteroid capture problem. However, some differences between them should be noted.

Based on a survey of the asteroid capture problem, we find that although there are different kinds of terminal orbits for asteroid missions, such as the Sun–Earth LPOs (Ceriotti & Sanchez 2016; Sánchez & Yárnoz 2016; Tang & Jiang 2016), the Earth–Moon LPOs (Tan et al. 2017), the distant retrograde orbit around the Moon (Strange et al. 2013), the distant prograde orbit around the Earth (Mingotti, Sánchez & McInnes 2014), the bounded Earth orbits (Bao et al. 2015) and so on, their basic goals are consistent, i.e. 'being captured' by the EMS. However, the WSB theory can only solve the gravitational capture problem in the region near a specific celestial body, such as the Moon, Mars, or Mercury, rather than a planet–moon binary system, such as the EMS.

The capture phenomena are common in planet–moon systems, in particular, the comet capture problem in the Jovian system has been well studied (Urrutxua & Bombardelli 2017). However, in the EMS, only asteroid 2006 RH_{120} was observed to be captured temporarily from 2006 September to 2007 June. Granvik, Vaubaillon & Jedicke (2012) first proposed a concept of TC to describe and investigate this phenomenon in the EMS. Temporarily-captured orbiters (TCOs) are defined as those which have a negative planetocentric energy while orbiting within a planetocentric distance of three Hill radii ($\sim$0.03 au in case of the Earth) during an extended but finite period of time. Based on their computation, at any given time there should be at least one temporarily-captured irregular natural Earth satellite of 1 m diameter orbiting the Earth. After that, taking asteroid 2006 RH_{120} as an example, Urrutxua et al. (2015) investigated how to extend the duration of such TC phases by acting upon the asteroid with slow deflection techniques that conveniently modify their trajectories. However, a recent study of Urrutxua & Bombardelli (2017) indicated that the definition of the TCO was defective. In their paper, they gave an example of an asteroid staying in the region of the EMS for a long time is wrongly classified as a temporarily-captured flyby (TCF), because its orbit about the Earth combines retrograde and prograde motions, such that the aggregated count of revolutions lies below unity. However, an asteroid escaping from the region of the EMS directly is labelled as a TCO, because its trajectory completes a full revolution around the Earth in the synodic *XY* plane, whereas a front view would reveal that the trajectory's projection in the synodic *XZ* plane does not even complete a full revolution.

Based on the discussion above, in this paper, a new concept called short-term capture (STC) is proposed and investigated to address the asteroid capture problem in the EMS. In the Sun–Earth–Moon–asteroid restricted four-body problem (RFBP), the occurrence conditions of the STC, including a space condition and a time condition, will be analysed. Then, the potential asteroids satisfying the STC conditions are found and studied. It should be noted that in this paper, the orbital motions of the Sun, Earth, Moon, and asteroid are assumed to lie in the same plane. This assumption is reasonable because Granvik et al. (2012) pointed out that the likelihood of a close Earth encounter is highest for Earth-like orbits, which have the heliocentric orbital elements: semimajor axis $a_s \sim 1$ au, eccentricity $e_s \sim 0$, and inclination $i_s \sim 0$.

According to the discussion above, the structure of this paper can be divided into six parts. In Section 2, the definition of the STC is proposed and compared with those of the WSB and the TC. In Section 3, the space condition of the STC is preliminarily analysed. In Section 4, the time condition of the STC is investigated in the planar RFBP. In Section 5, the probabilities of the STC and its index are proposed to analyse the influence of parameters on the STC. In Section 6, potential STC asteroids are found and discussed. Finally, the conclusions of this paper are presented in Section 7.

2 DEFINITION FOR SHORT-TERM CAPTURE

In this section, the definitions of the WSB and the TC are first discussed. Then, the definition of STC in the EMS is given.

Belbruno (1987) first proposed the theory of the WSB to design fuel efficient space missions, but this initial definition was not very rigorous. After that, many scholars investigated the WSB and gradually refined the definition. To the authors' best knowledge, the most recent and rigorous algorithmic definition of the WSB was presented in Belbruno, Gidea & Topputo (2010).

In the planar circular restricted three-body problem (PCRTBP), the larger and smaller primaries are denoted by P_1 and P_2, respectively. We consider trajectories of the infinitesimal particle P_3 around P_2 with the following initial conditions (Belbruno et al. 2010):

(i) The initial position of the trajectory is on a radial segment $l(\theta)$ in the configuration space departing from P_2 and making an angle of θ with the P_1P_2 line, relative to the rotating system. The trajectory is assumed to start at the periapsis of an osculating ellipse around P_2, whose semimajor axis lies on $l(\theta)$ and whose eccentricity e is held fixed along $l(\theta)$. The initial velocity of the trajectory is perpendicular to $l(\theta)$; there are two different such choices of initial velocities, one positive (direct motion) and one negative (retrograde motion).

(ii) The initial Keplerian energy H_2 relative to P_2 is negative; i.e. $H_2 < 0$.

(iii) Then, the motion is said to be n-stable if the infinitesimal mass P_3 leaves $l(\theta)$ and makes n complete turns about P_2 without making a complete turn around P_1, and if the intersections of the trajectory with $l(\theta)$ along this trajectory are all transverse intersections and have negative Kepler energy with respect to P_2 (i.e. $H_2 < 0$). The motion is otherwise said to be n-unstable.

In our opinion, the three conditions above can be summarized in three keywords: (1) periapsis around P_2, (2) negative Keplerian energy H_2, and (3) the complete turn with negative Keplerian energy. A weak stable capture can be achieved if the above three conditions are met. Note that the weak stable capture is a TC rather than a permanent capture, P_3 will escape from the region of P_2 after several turns if a braking manoeuvre is not performed.

Granvik et al. (2012) proposed a twofold definition for temporary satellite capture by a planet (or any object orbiting the Sun including the Moon) by requiring simultaneously

(i) The planetocentric Keplerian energy $H_2 < 0$.

(ii) The planetocentric distance is less than three Hill radii for the planet in question (e.g. for the Earth 0.03 au).

The TCO can make at least one full revolution around the planet in a Sun–planet rotating frame while being captured, but the TCF fails to complete a full revolution around the planet.

As we can see from the above definition, the conditions of a TCO can also be summarized by three keywords: (1) negative Keplerian energy H_2, (2) three Hill radii, and (3) one full revolution around the planet in a rotating frame. Comparing the definitions of the WSB and the TCO, they actually are quite similar. However, as mentioned

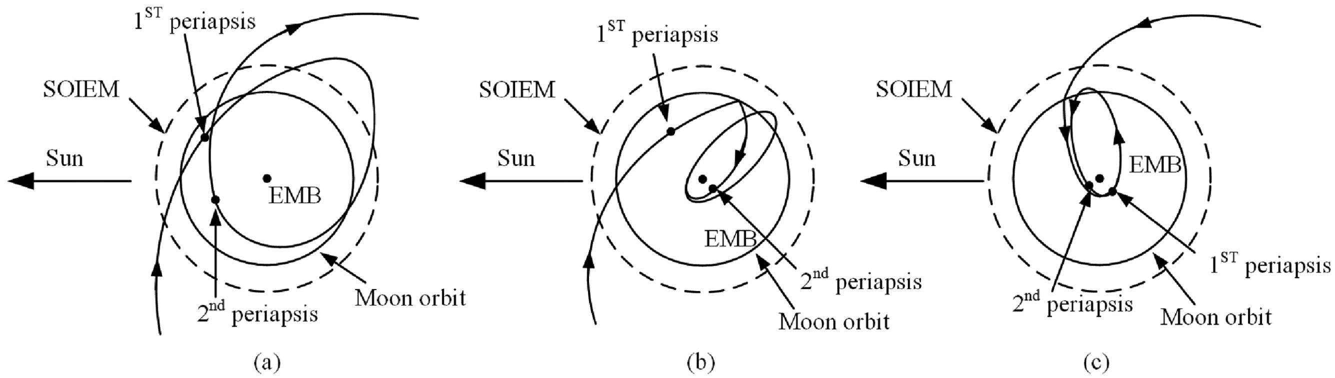

Figure 1. Some sketches of the STC orbits in the Sun–EMB rotating frame.

in Section 1, the above two theories are both defective in the capture problem in the EMS.

As mentioned before, the WSB can only address the capture problem in the region of a specific celestial body, such as the Moon, Mars, or Mercury, but cannot directly be applied to the capture problem in a planet–moon binary system. A feasible alternative is to use the Sun–Earth/Moon CRTBP where the Earth and the Moon are combined into a single primary located at the Earth–Moon barycentre (EMB) with the combined mass of the Earth and the Moon, instead of the Sun–Earth–Moon RFBP. In this way, the weak stable capture around the EMB can be analysed by the WSB theory. However, according to Qi & Xu (2016), the Jacobi constant C_{SE} of the Sun–Earth/Moon CRTBP can be changed significantly by the Moon inside the sphere of influence of the EMS (SOIEM), which is a disc region centred at the EMB with radius of $r_{\mathrm{soiem}} = 579\,734.2$ km, so the WSB may be invalidated inside the SOIEM. Based on the analysis of the SOIEM (Qi & Xu 2016), inside the SOIEM, the Earth–Moon CRTBP can approximate the dynamics instead of the Sun–Earth–Moon RFBP, while outside the SOIEM, the Sun–Earth–Moon RFBP can be approximately replaced by the Sun–Earth/Moon CRTBP. Yegorov (1961), Huang & Innanen (1983), and Tanikawa (1983) indicated that, just using the pure gravitational CRTBP, the orbiter can only be temporarily captured. If the orbiter is required to insert into a long-term capture orbit, a braking manoeuvre is necessary (Urrutxua et al. 2015). To reduce the fuel cost, we require the trajectory to pass through the SOIEM, because only inside the SOIEM can the LGA decrease the energy and velocity of the orbiter efficiently (Qi & Xu 2015). In addition, Granvik et al. (2012) indicated that although conventional wisdom suggests that the Moon's orbit is the primary reason for the lack of long-lived TCOs, it is simultaneously the presence of the Moon that increases the capture probability by allowing faster objects to be captured. Therefore, we postulate that the SOIEM is very important to the capture problem in the EMS. To make sure that the asteroid's orbit can pass through the SOIEM, we demand that its periapsis with respect to the EMB should be located inside the SOIEM. We also require that the capture orbit cannot escape from the SOIEM and directly insert into a heliocentric orbit. The definitions of the WSB and the TCO both require a complete turn or revolution around the planet in a rotating frame. By comparison, the STC also requires at least one complete revolution around the EMB but not in a rotating frame. Urrutxua & Bombardelli (2017) found that an asteroid staying in the region of the EMS for a long time may be wrongly classified as a TCF, because its orbit combines retrograde and prograde motions, such that the aggregated count of revolutions lies below unity. However, an asteroid escaping from the region of EMS directly is labelled as a TCO, because its trajectory completes a full revolution around the Earth in the synodic *XY* plane, whereas a front view would reveal that the trajectory's projection in the synodic *XZ* plane does not even complete a full revolution. Therefore, they pointed out that the procedure for counting revolutions proposed by Granvik et al. (2012) may not be appropriate. We postulate that the revolutions measured in the rotating frame are the source of the problem. In this paper, one complete revolution or turn is defined as the portion between two consecutive periapsides with respect to the EMB, i.e. one revolution with respect to the EMB in the EMB-centric inertial frame. In this paper, we require that the STC trajectory makes at least one revolution inside the SOIEM or returns to the SOIEM after the first flyby of the SOIEM. If there are at least two periapsides with respect to the EMB located inside the SOIEM, we find that both requirements of the STC orbit as given above can satisfied.

In addition, the STC also requires that the asteroid comes from the region outside the EMS. In this paper, we define the region of the EMS as a disc region centred at the EMB with radius of three Hill radii for the Earth (∼0.03 au), similar to the TC proposed by Granvik et al. (2012).

In summary, we propose a twofold definition for a STC in the EMS by requiring the following:

(i) the orbiter comes from the region outside the EMS and enters the SOIEM, and then

(ii) at least two periapsides with respect to the EMB are located inside the SOIEM before the orbiter leaves the region of the EMS.

Note that in the above definition of the STC, the requirement of the Keplerian energy in the WSB and the TC is abandoned, because the EMB-centric Keplerian energy can well reflect the orbital characters around the EMB in the Sun–Earth/Moon CRTBP, but cannot completely and properly describe the orbital characters in the EMS where the Earth and Moon are treated as separate celestial bodies. For example, when the asteroid passes by the Moon, its EMB-centric Keplerian energy will fluctuate dramatically (Qi & Xu 2015).

Fig. 1 displays some sketches of the STC orbits in the Sun–EMB rotating frame, where the LGA plays an important role in (b) and (c). Fig. 2 displays the detail of the trajectory passing through the SOIEM. The geometric information of the orbit entering and escaping the SOIEM can be denoted by (α_1, β_1) and (α_2, β_2), respectively (Qi & Xu 2016). θ_{m0} indicates the initial phase angle of the Moon when the orbit enters the SOIEM. If the orbit passes through the Moon orbit, the phase angles of two cross points, C1 and C2, are denoted by θ_{c1} and θ_{c2}, respectively.

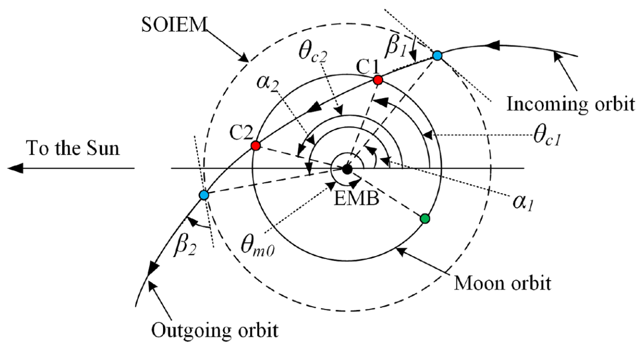

Figure 2. Detail of the orbit inside the SOIEM.

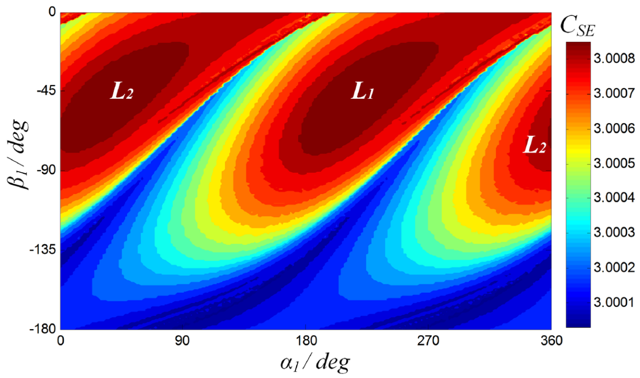

Figure 3. Poincare sections of unstable manifolds in the SOIEM with different $C_{\rm SE}$.

Qi, Xu & Qi (2017) proposed the Poincare sections of the unstable manifolds associated with the LPO of the Sun–Earth/Moon CRTBP on the SOIEM, described by α_1 and β_1 (see Fig. 2). According to their research, for a given Jacobi constant in the Sun–Earth/Moon CRTBP $C_{\rm SE}$, only the points inside the Poincare sections corresponding to $C_{\rm SE}$ can escape from the SOIEM to the region outside the EMS by backward integration. Hence, the regions inside the Poincare sections corresponding to $C_{\rm SE}$ are the feasible regions for condition 1 of the STC, which are denoted by $U_c(C_{\rm SE})$. Fig. 3 shows the distributions of the Poincare sections of the unstable manifolds with different $C_{\rm SE}$, where the colours denote the magnitude of $C_{\rm SE}$. As we can see from the figure, the smaller $C_{\rm SE}$, the larger $U_c(C_{\rm SE})$ is. This result can easily be explained: a smaller $C_{\rm SE}$ corresponds to a larger velocity in the Sun–EMB rotating frame, so it is more likely to escape from the SOIEM by backward integration.

In this paper, the model considered for the Sun–Earth–Moon RFBP is the planar bicircular model (PBCM). The PBCM assumes that the Earth and the Moon are revolving in circular orbits around the EMB; meanwhile, the Sun and the EMB are also assumed in circular orbits around their common centre of mass. For the detail and the equations of motion of the PBCM, readers can refer to Qi & Xu (2016). In the PBCM, a capture orbit can be uniquely specified by four parameters as follows (see Fig. 2):

(i) $C_{\rm SE}$, the Jacobi constant of the Sun–Earth/Moon CRTBP when the asteroid first enters the SOIEM.

(ii) α_1, the phase angle of the entry point when the asteroid first enters the SOIEM.

(iii) β_1, the angle of the velocity when the asteroid first enters the SOIEM.

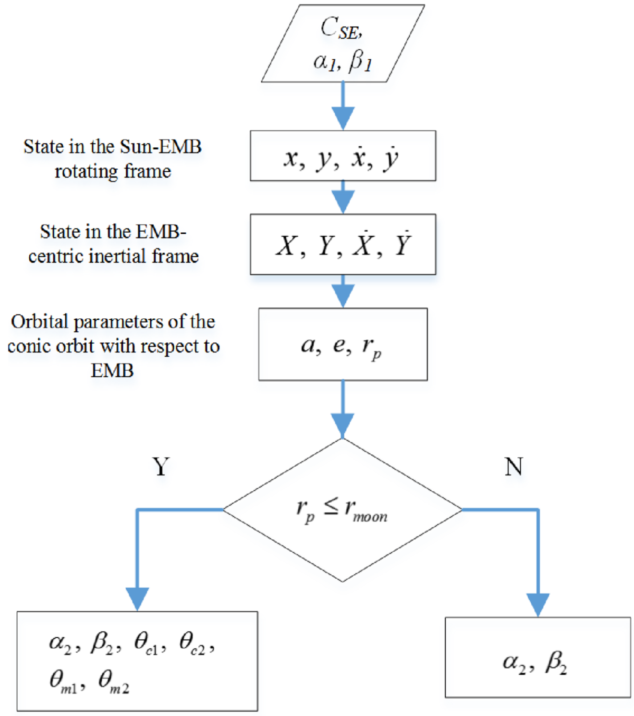

Figure 4. Calculation process for the angles in the SOIEM.

(iv) θ_{m0}, the phase angle of the Moon when the asteroid first enters the SOIEM.

$C_{\rm SE}$ determines the feasible region U_c. α_1 and β_1 determine the initial state of the orbit on the SOIEM, while θ_{m0} determines the initial moment on the SOIEM. For a given $C_{\rm SE}$, we first analyse the space condition of the STC in (α_1, β_1) coordinates, then the time condition of the STC will be examined by scanning θ_{m0}.

3 PRELIMINARY ANALYSIS: SPACE CONDITION

In this section, the influence of the Moon is neglected, so the PBCM is simplified into the Sun–Earth/Moon CRTBP to perform a preliminary analysis of the space condition of the STC. Furthermore, the Sun–Earth/Moon CRTBP inside the SOIEM is approximately replaced by the EMB-centric two-body model.

We assume that the orbit inside the SOIEM can be approximately described by a conic orbit in the EMB-centric two-body model. Based on the classical formulas of the two-body model (Chobotov 2002), we can calculate (α_2, β_2) when the asteroid leaves the SOIEM via $C_{\rm SE}$, α_1, and β_1. If the orbit passes through the Moon orbit, the phase angles θ_{c1} and θ_{c2} can also be obtained. The process is displayed in the flow chart in Fig. 4, where θ_{m1} is the initial phase angle of the Moon when the orbit first enters the SOIEM if the asteroid and the Moon arrive at C1 simultaneously. In the same way, θ_{m2} can also be defined if the asteroid and the Moon arrive at C2 simultaneously. $r_{\rm moon}$ is the radius of the Moon orbit in the Sun–EMB rotating frame, 3.7973 × 10^5 km. a, e, and r_p are the semimajor axis, eccentricity, and the distance of periapsis from the EMB, respectively. The details of the calculations can be found in Appendix A.

In the EMB-centric two-body model, we can prove $\beta_2 = -\beta_1$ (see the details in Appendix A). In the Sun–Earth/Moon CRTBP, $C_{\rm SE}$ is invariant in the SOIEM. Hence, the parameters of the orbit

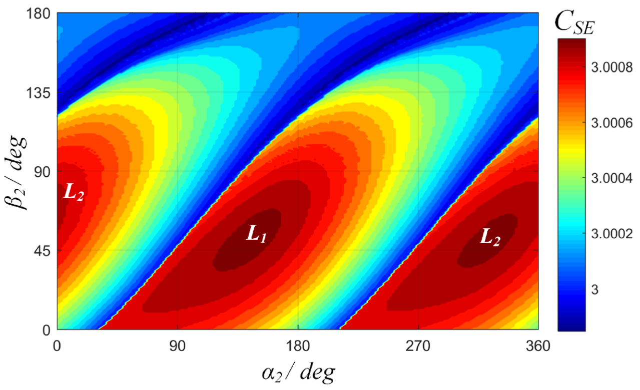

Figure 5. Poincare sections of stable manifolds in the SOIEM with different C_{SE}.

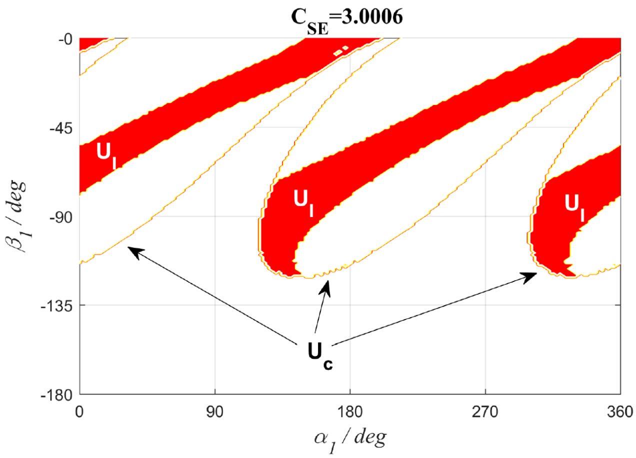

Figure 7. Distributions of U_I and U_c with $C_{\mathrm{SE}} = 3.0006$.

leaving the SOIEM, including C_{SE}, α_2, and β_2, can be obtained. Qi & Xu (2016) pointed out that for a given C_{SE}, we can use (α_2, β_2) to determine the motion of the orbit: if (α_2, β_2) is located inside the Poincare sections of the stable manifolds of the LPO corresponding to C_{SE} on the SOIEM (see Fig. 5), the orbit will pass through the region of the libration point and will leave the region of the EMS, otherwise, the orbit will return to the SOIEM in the sense of probability. For example, Fig. 6 shows the Poincare sections of unstable manifolds (left figure) and the stable manifolds (right figure) when $C_{\mathrm{SE}} = 3.000\,85$. Based on the analysis above, if (α_1, β_1) inside the Poincare section of the left figure corresponds to a (α_2, $-\beta_1$) outside the Poincare section of the right figure (see the blue dashed line in Fig. 6), the orbit can not only come from the outside region of the EMS but also return to the SOIEM after the first flyby of the SOIEM, that is to say, the orbit can satisfy the requirements of the STC. On the contrary, if (α_2, $-\beta_1$) is located inside the Poincare section of the right figure (see the red line in Fig. 6), the orbit will directly leave the SOIEM through the invariant manifold tube after the first flyby of the SOIEM (Qi & Xu 2016).

Using the above method, for a given C_{SE}, we can test any (α_1, β_1) $\in U_c$ and find the points satisfying the requirements of the STC. In this way, the space condition of the STC translates to a space constraint in (α_1, β_1) coordinates. The set satisfying the requirements of the STC in (α_1, β_1) coordinates is defined as U_I. Apparently, $U_I \subset U_c$. Fig. 7 displays the distributions of U_c and U_I when $C_{\mathrm{SE}} = 3.0006$.

Based on the discussion thus far, U_I can realize STCs via the dynamical theory of the Sun–Earth/Moon CRTBP without considering the influence of the Moon. If we take the LGA into consideration, we can find other space conditions for STC. According to Gong & Li (2015), the LGA can change the magnitude of C_{SE} as much as 0.0015 approximately, so an initial C_{SE} larger than 2.9985 on the SOIEM can be increased to 3 or higher by the LGA. From Fig. 3, when C_{SE} is larger than 3, there exist regions outside the Poincare section, i.e. regions returning to the SOIEM. Therefore, if an asteroid passes through the Moon orbit, it will have a chance to pass by the Moon in a close distance and realize a STC.

The condition to pass through the Moon orbit can also be found by using the EMB-centric two-body model in the SOIEM. Based on the method in Fig. 4, for a given C_{SE}, (α_1, β_1) corresponding to $r_p \le r_{\mathrm{moon}}$ can be identified. In this way, the space condition of the LGA can also translate to a space constraint in (α_1, β_1) coordinates. We define the set satisfying $r_p \le r_{\mathrm{moon}}$ in (α_1, β_1) coordinates as U_{II}. For a given C_{SE}, (α_1, β_1) on the boundary of U_{II} corresponds to $r_p = r_{\mathrm{moon}}$. Hence, for a given (α_1, β_1), we define C_{SE} satisfying $r_p = r_{\mathrm{moon}}$ as C^*_{SE}. Fig. 8 illustrates the distribution of C^*_{SE} in (α_1, β_1) coordinates. As we can see from the figure, the distribution of C^*_{SE} gradually moves to the centre from the upper and lower boundaries with the decrease of C^*_{SE}. Using the geometric relationship, we can determine that the distribution of C^*_{SE} tends to the lines $\beta_1 = -\arccos(r_{\mathrm{moon}}/r_{\mathrm{soiem}}) \approx -49.08^\circ$ and $\beta_1 = \arccos(r_{\mathrm{moon}}/r_{\mathrm{soiem}}) - 180^\circ \approx -130.92^\circ$ when C_{SE} tends to negative infinity. For a given

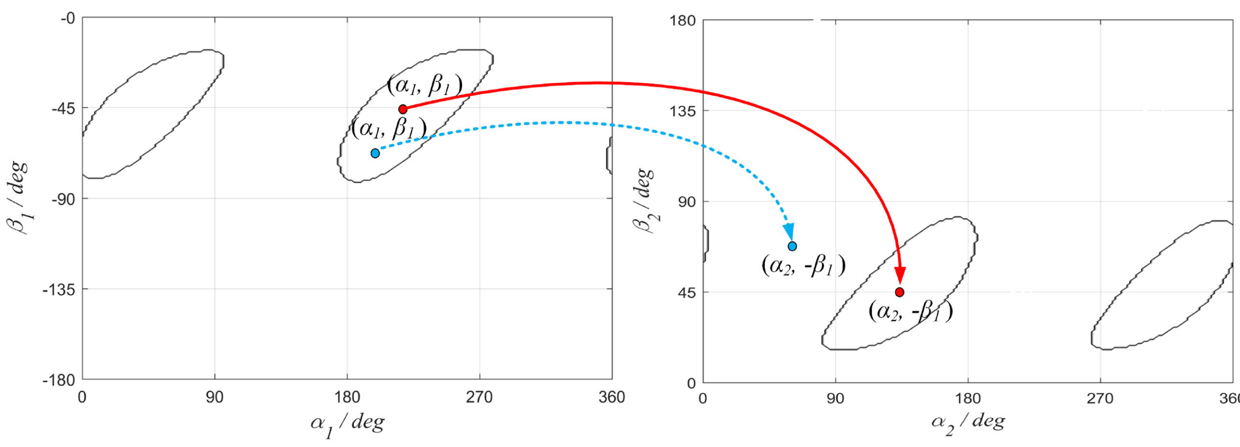

Figure 6. Determination method of the motion using the Poincare sections of the invariant manifolds.

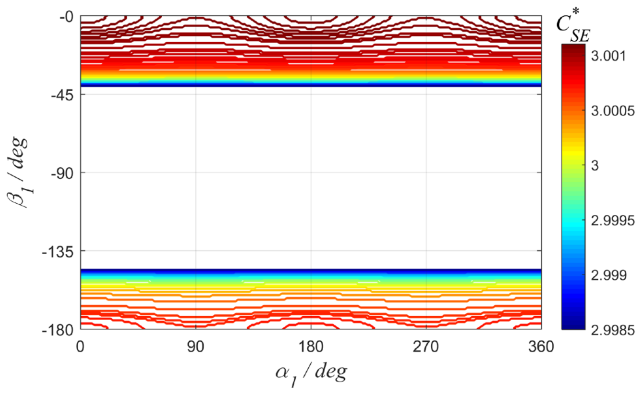

Figure 8. Distribution of C^*_{SE}.

(α_1, β_1), if its C_{SE} is smaller than C^*_{SE} at this point, the orbit cannot pass through the Moon orbit, otherwise, it can. Therefore, as we can see from Fig. 8, U_{II} gradually shrinks with the decrease of C_{SE}. The region between -49.08° and -130.92° can satisfy the condition to pass through the Moon orbit for any C_{SE}.

Besides the above two space conditions, another condition should be also noted, i.e. the orbit cannot pass through the Earth. Technically, r_p is the distance between periapsis and the EMB rather than the Earth's centre, but the distance between the EMB and the Earth's centre in the Sun–EMB rotating frame is quite small and even smaller than the radius of the Earth, $r_{\mathrm{earth}} = 6378$ km. For simplicity, we directly demand $r_p > r_{\mathrm{earth}}$. The region corresponding to $r_p \leq r_{\mathrm{earth}}$ in (α_1, β_1) coordinates is defined as U_e. Apparently, $U_e \subset U_{II}$. Numerical computation indicates that U_e is approximately distributed in the region where $\beta_1 \in [-105, -90^\circ]$. Based on the discussion thus far, for a given C_{se}, the feasible region U_c can be divided into five subsets as follows:

(i) $U_1 = U_I \cap (U_c - U_{II})$, i.e. the orbit can realize the STC directly but cannot pass through the Moon orbit.
(ii) $U_2 = U_I \cap (U_{II} - U_e)$, i.e. the orbit can not only realize the STC directly but also pass through the Moon orbit. In addition, the orbit will not impact the Earth.
(iii) $U_3 = (U_{II} - U_e) \cap (U_c - U_I)$, i.e. the orbit cannot realize the STC directly but can pass through the Moon orbit. In addition, the orbit will not impact the Earth.
(iv) $U_4 = U_e$, i.e. the orbit will impact the Earth.
(v) $U_5 = U_c - (U_I \cup U_{II})$, i.e. the orbit cannot realize the STC directly or pass through the Moon orbit.

Fig. 9 displays the distributions of the five subsets with $C_{\mathrm{SE}} = 3.0006$ and five corresponding orbits in the Sun–EMB rotating frame. The outer circles and inner circles in these figures are the SOIEM and the Moon orbits, respectively. The trajectories are obtained by numerical integration in the Sun–Earth/Moon PCRTBP.

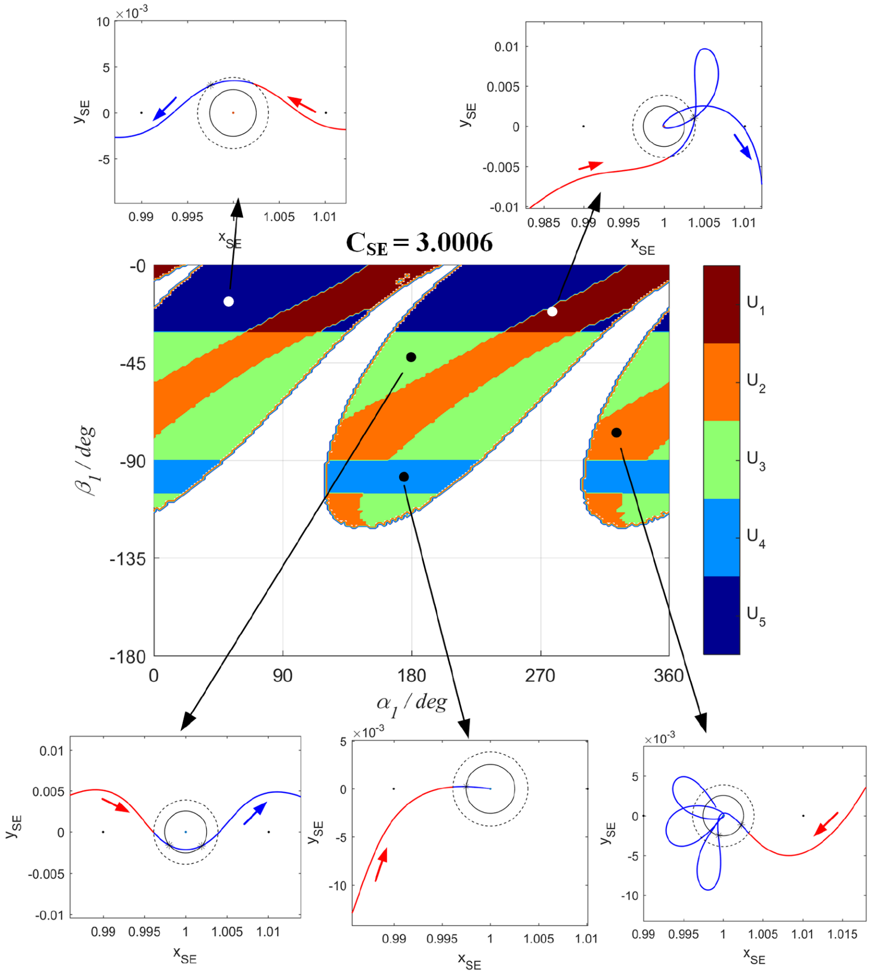

Figure 9. Distributions of five subsets with $C_{\mathrm{SE}} = 3.0006$ and five corresponding orbits.

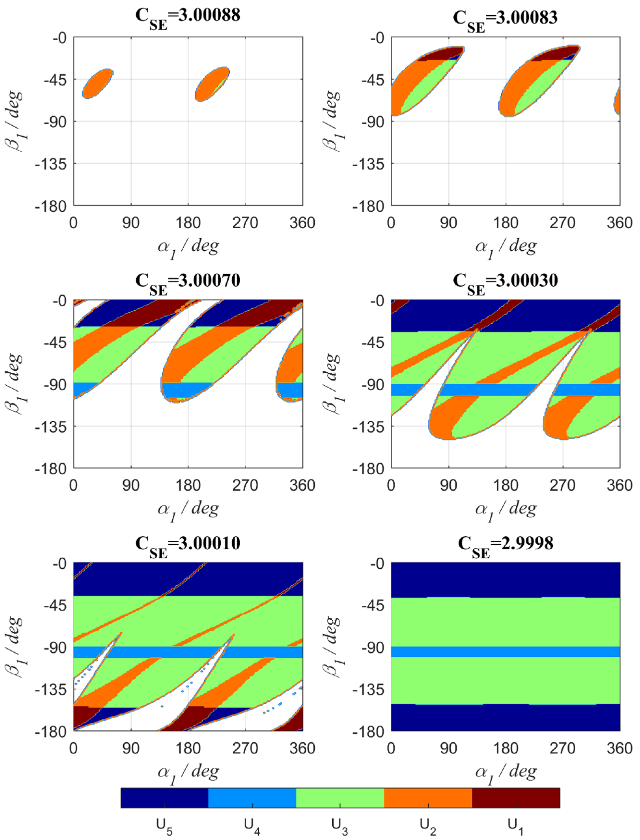

Figure 10. Distributions of five subsets with different C_{SE}.

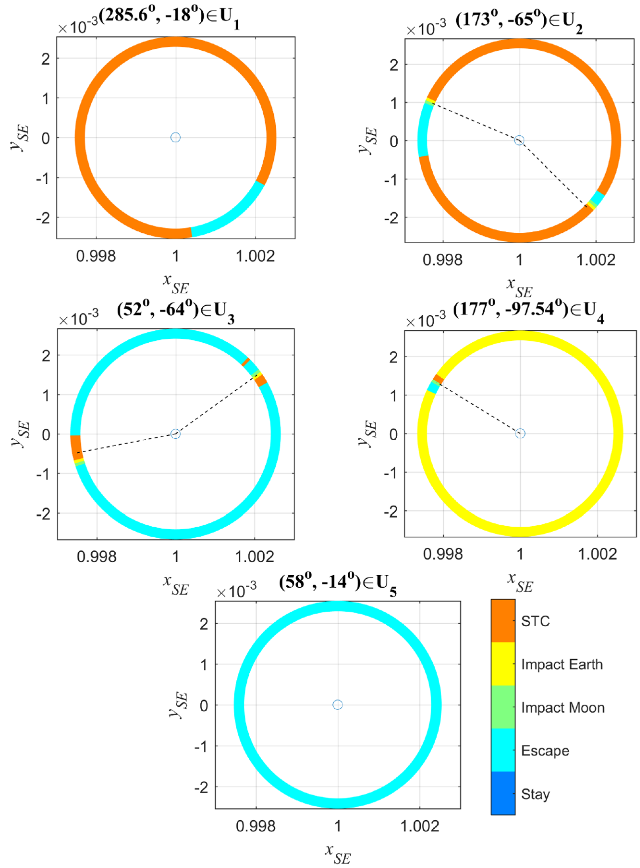

Figure 11. Distribution of five outcomes for different θ_{m0} and subsets when $C_{\mathrm{SE}} = 3.0006$.

Fig. 10 shows the distributions of the subsets with different C_{SE}. In these figures, the region of U_c becomes larger with the decrease of C_{SE}. Numerical computations indicate that when C_{SE} is smaller than 2.9999, the feasible region U_c is the whole region in (α_1, β_1) coordinates, and U_1 and U_2 both disappear, so only via a LGA may STC be realized. When $C_{\mathrm{SE}} = 3.000\,88$, as we can see, almost all orbits in U_c can achieve the STC directly by the dynamical theory of the Sun–Earth/Moon PCRTBP, meanwhile all of them pass through the Moon orbit.

It should be noted that above preliminary analysis and result are based on the Sun–Earth/Moon PCRTBP and EMB-centric two-body model, where the influence of the Moon is not taken into consideration. Therefore, only the space condition of the STC is obtained. The influence of the Moon or a LGA actually is a double-edged sword for STC: on one hand, some orbits in the subsets U_3 and U_4 may achieve STC by a LGA; on the other hand, some orbits in the subsets U_1 and U_2 may escape from the region of the EMS directly due to a LGA. Hence, a further analysis about STC should take the Moon into consideration. In the SOIEM, the influence of the Moon depends on the relative position between the orbit and the Moon, which is related to the phase angle of the Moon when the orbiter first enters the SOIEM, θ_{m0}. In the next section, we will discuss the time condition of the STC in the Sun–Earth–Moon PBCM by scanning θ_{m0}.

4 ANALYSIS IN THE PBCM: TIME CONDITION

As mentioned previously, θ_{m0} determines the time condition of STC. For the five kinds of subsets proposed in the previous section, we investigate the motions of the corresponding orbits by scanning θ_{m0} in the PBCM, respectively. Inside the region of the EMS, after the asteroid first enters the SOIEM, five possible outcomes of the orbit exist: the asteroid achieves STC; the asteroid impacts the Earth; the asteroid impacts the Moon; the asteroid escapes the region of the EMS directly; the asteroid stays in the region of the EMS but does not return to the SOIEM during the integration time. In this paper, the above five outcomes are labelled as 'STC', 'Impact Earth', 'Impact Moon', 'Escape', and 'Stay', respectively.

Taking the feasible region U_c with $C_{\mathrm{SE}} = 3.0006$ as an example, we choose five points from the five subsets $U_1 \sim U_5$ as the initial state of the numerical integration in the PBCM, respectively. Then, 200 θ_{m0}s are evenly sampled from the complete value range of θ_{m0}, $[0,\ 2\pi]$. The numerical integration time is 10 time units of the Sun–Earth/Moon CRTBP, about 581.27 d. By scanning θ_{m0}, the outcomes of the five points in U_c with different θ_{m0} are obtained in the PBCM. Fig. 11 displays the numerical results in the Sun–EMB rotating frame, where the circles are the Moon orbits. Since the orbits in U_2 and U_3 pass through the Moon orbit, their θ_{m1} and θ_{m2} can be obtained using the computation method in the previous section, and are displayed in Fig. 11 by the dashed lines. The orbits in U_4 enter the Moon orbit but impact the Earth, so only θ_{m1} is meaningful, and is indicated by the dashed line.

The orbit corresponding to the point $(285.6^\circ, -18^\circ) \in U_1$ can achieve STC by the dynamical system of the Sun–Earth/Moon PCRTBP without the influence of the Moon, so as we can see from the figure, for most of the time, STC can be realized. The only

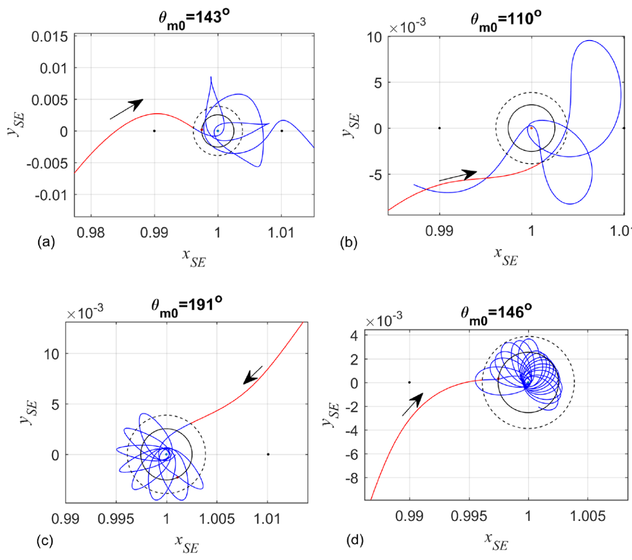

Figure 12. Four kinds of STC orbits when $C_{\rm SE} = 3.0006$.

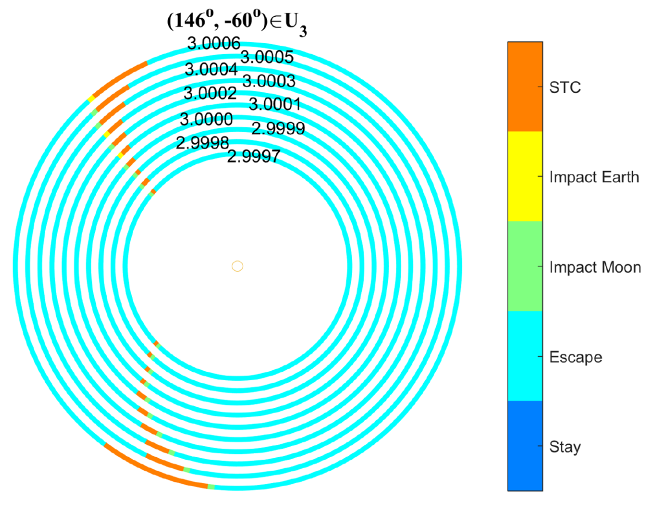

Figure 13. Distributions of the STC window for different $C_{\rm SE}$.

exception is the portion when the orbit passes by the Moon. Due to the LGA, the orbit can escape from the region of the EMS.

The orbit corresponding to the point $(173°, -65°) \in U_2$ can not only achieve STC without the influence of the Moon but also passes through the Moon orbit. By looking at the figure, we find that θ_{m1} and θ_{m2} obtained in the two-body model are very close to the portion of 'Impact Moon', indicating that the calculation method proposed in the two-body model can well approximate the orbit of the PBCM inside the SOIEM. Based on the property of the LGA, when lunar flyby occurs behind the Moon, the energy of the orbiter increases; when lunar flyby occurs in front of the Moon, the energy of the orbiter decreases (Qi & Xu 2015). The motion of the Moon is anticlockwise (or prograde) therefore we infer that the portions of 'Impact Earth' and 'Impact Moon' are usually located in the transition portion from the 'STC' to 'Escape' along the anticlockwise direction, and the portions of 'Stay' are usually located in the transition portion from the 'Escape' to 'STC' along the anticlockwise direction. Consequently, the portions of 'Escape' are approximately located in front of θ_{m1} and θ_{m2} along the anticlockwise direction, and the portions of 'STC' are approximately located behind θ_{m1} and θ_{m2} along the anticlockwise direction, which can be seen from Fig. 11.

The orbit corresponding to the point $(52°, -64°) \in U_3$ cannot achieve STC without a LGA but does pass through the Moon orbit. As we can see from the figure, for most of the time, the orbit will escape the region of the EMS directly, but in the narrow portions near θ_{m1} and θ_{m2}, STC can be realized by a LGA.

The orbit corresponding to the point $(177°, -97.54°) \in U_4$ will directly impact the Earth without LGA. Therefore, as we can see from the figure, the portion 'Impact Earth' dominates, and other portions can only be distributed in the narrow portion near θ_{m1}.

The orbit corresponding to the point $(58°, -14°) \in U_5$ will escape the region of the EMS directly in the Sun–Earth/Moon PCRTBP. Because the orbit cannot pass through the Moon orbit, the influence of the Moon on the orbit is quite slight. Hence, there are no other portions except 'Escape'.

Based on Fig. 11, four STC trajectories are obtained by numerical integration in the PBCM and displayed in Fig. 12, where the outer and inner circles in these figures are the SOIEM and the Moon orbits, respectively. The geometric states (α_1, β_1) of the trajectories from (a) to (d) correspond to the four points from U_1 to U_4 in Fig. 11, respectively. θ_{m0}s in Fig. 12 are chosen from the corresponding 'STC' portions in Fig. 11. As we can see, the orbits in (c) and (d) achieve STC due to a LGA. According to existing research about the LGA (Gong & Li 2015; Qi & Xu 2015), a lunar flyby closer to the Moon can generate a larger energy change of the orbiter. Therefore, when θ_{m0} is located in the 'STC' portion near θ_{m1} or θ_{m2}, the LGA can significantly reduce the energy or velocity of the orbit, resulting in the capture orbit being constrained in the region near the Earth, such as the orbits in (c) and (d) in Fig. 12.

In this paper, the 'STC' portions in the range of θ_{m0}, $[0, 2\pi]$, are defined as the STC windows. From Fig. 11, the orbits in U_1 and U_2 have wider STC windows; the STC windows of U_3 and U_4 are narrow; U_5 has no STC window. Considering that the portions of 'STC' and 'Escape' usually exist in a neighbourhood of θ_{m1} and θ_{m2}, we postulate that the STC windows of U_2 are usually smaller than those of U_1, and similarly, the STC windows of U_4 are usually smaller than those of U_3. Therefore, we infer that, the STC window gradually shrinks from U_1 to U_5. As demonstrated in Fig. 10, with the decrease of $C_{\rm SE}$, the subset U_3 will gradually dominate in the feasible region U_c. Since most NEAs have small $C_{\rm SE}$ (Sánchez & Yárnoz 2016), we need to pay more attention to the STC windows of U_3. Fig. 13 shows the distributions of 'STC' portions with $(146°, -60°) \in U_3$ and different $C_{\rm SE}$. As we can see, the STC window gradually shrinks with the decrease of $C_{\rm SE}$.

From the STC window, the time probability of STC, η_t, can be naturally defined as the ratio of the length of the STC window to the length of the range of θ_{m0},

$$\eta_t = \theta_{\rm stc}/2\pi, \tag{1}$$

where $\theta_{\rm stc}$ is the length of the STC window. From Figs 11 and 13, $\theta_{\rm stc}$ is determined by $C_{\rm SE}$, α_1, and β_1, so η_t is a function of $C_{\rm SE}$, α_1, and β_1, i.e. $\eta_t(C_{\rm SE}, \alpha_1, \beta_1)$. Fig. 14 displays the distributions of η_t with different $C_{\rm SE}$ in (α_1, β_1) coordinates. By looking at the figure, we find some interesting phenomena: the distributions of η_t obtained in the PBCM are similar to those of the five subsets proposed in the Sun–Earth/Moon PCRTBP and the EMB-centric two-body model (see Fig. 10). Comparing the results in Figs 10 and 14, we find that η_ts larger than 0.6 are generally located in U_1 and U_2, in particular η_ts in U_1 are larger than 0.8; η_ts in U_3 are relatively small (<0.6) and seem to decrease with the decrease of $C_{\rm SE}$, similar to the conclusion in Fig. 13; η_ts in U_4 are generally

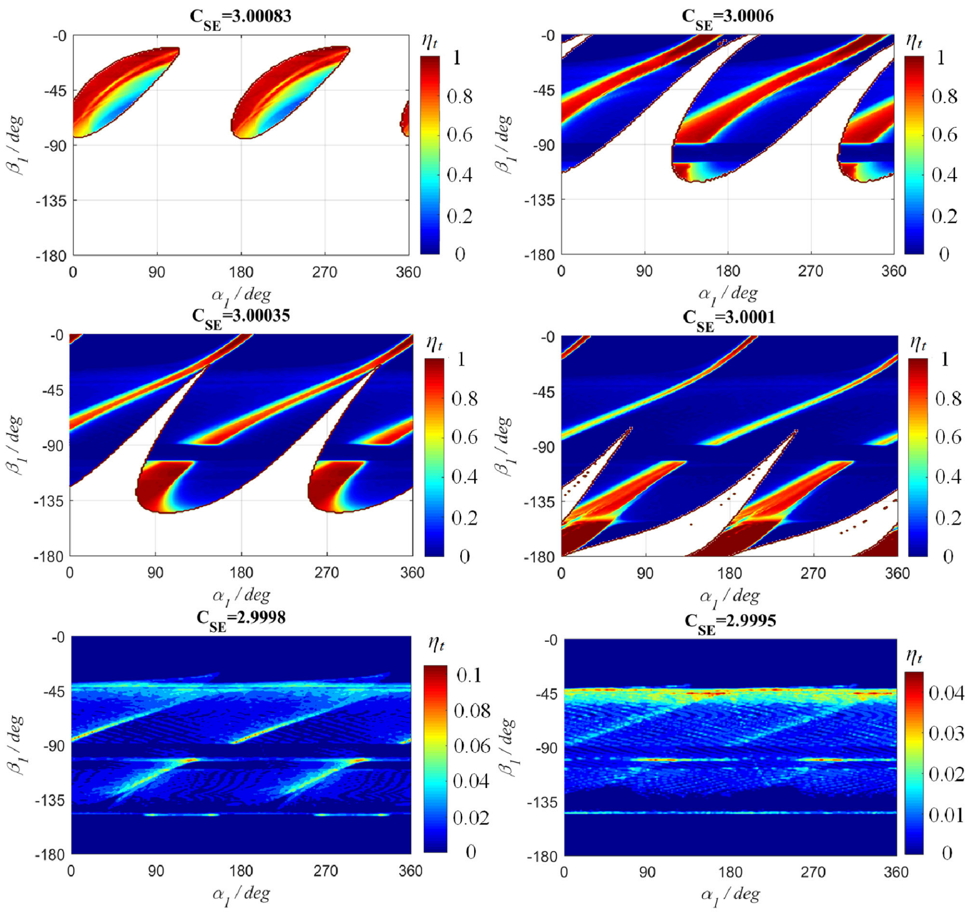

Figure 14. Distributions of η_t for different C_{SE}.

smaller than 0.01; η_ts of U_5 are equal to 0 in most regions except the borders between U_5 and other subsets. Those results are consistent with our previous analysis about the STC windows for U_1–U_5 in Fig. 11, proving that our division of the five subsets is meaningful and useful although they are obtained from simple models and preliminary analysis. In this way, the space condition of STC in an autonomous system is connected with the time condition of STC in a time-varying system.

5 PROBABILITY OF STC

For a given C_{SE}, the relative space probability of the subset U_i, $i = 1, 2, \ldots, 5$, in the feasible region U_c is defined as the ratio of the measure of U_i to the measure of U_c,

$$\eta_{si} = m(U_i)/m(U_c), \qquad i = 1, 2, \ldots, 5, \tag{2}$$

where $m(\cdot)$ is the measure function of subset U_i. $m(U_i)$ is obtained by

$$m(U_i) = \iint\limits_{U_c} \chi_{ui} \mathrm{d}\alpha_1 \mathrm{d}\beta_1, \tag{3}$$

where χ_{ui} is the indicator function of subset U_i,

$$\chi_{ui}(\alpha_1, \beta_1) = \begin{cases} 1, & \text{if} \quad (\alpha_1, \beta_1) \in U_i; \\ 0, & \text{if} \quad (\alpha_1, \beta_1) \notin U_i. \end{cases} \tag{4}$$

Apparently, $m(U_c) = \sum_{i=1}^{5} m(U_i)$.

Based on the discussion above, because η_ts in U_4 and U_5 are generally smaller than 0.01, we consider that their STC window is too narrow to be suitable for STC. Hence, in this paper, only U_1–U_3 can be regarded as suitable regions for STC in U_c, and the space probability of $\bigcup_{i=1}^{3} U_i$ is defined as the relative space probability of STC, $\eta_s = \sum_{i=1}^{3} \eta_{si}$. In addition, the absolute space probability of STC in the full space of (α_1, β_1) coordinates, $\Omega = \{(\alpha_1, \beta_1) | \alpha_1 \in [0^\circ, 360^\circ], \beta_1 \in [-180^\circ, 0^\circ]\}$, is defined as

$$\eta_{sf} = \sum_{i=1}^{3} m(U_i)/m(\Omega), \tag{5}$$

Fig. 15 displays the curves of η_{si}, η_s, and η_{sf} with different C_{SE}. In this figure, the curve of η_{s3} overlaps those of η_s and η_{sf} when C_{SE} is smaller than 2.9999. By looking at the figure, we find that η_s generally increases with an increase in C_{SE}, but η_{sf} decreases when C_{SE} is larger than 3 due to the reduction of U_c. In Fig. 15, $C_{\mathrm{SE}} = 2.9999$ is an important dividing line for the curves. When C_{SE} is smaller than 2.9999, η_{s1} and η_{s2} decrease to 0, so only U_3 is suitable for STC, i.e. $\eta_s = \eta_{s3}$. In addition, when C_{SE} is smaller than 2.9999, the changes of the curves are quite slow, but when C_{SE} is larger than 2.9999, the curves fluctuate relatively dramatically.

According to the definitions of space probabilities, the mean time probability of STC for a given C_{SE} can be defined as

$$\bar{\eta}_t(C_{\mathrm{SE}}) = \frac{\iint\limits_{U_c} \eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1) \mathrm{d}\alpha_1 \mathrm{d}\beta_1}{m(U_c)}. \tag{6}$$

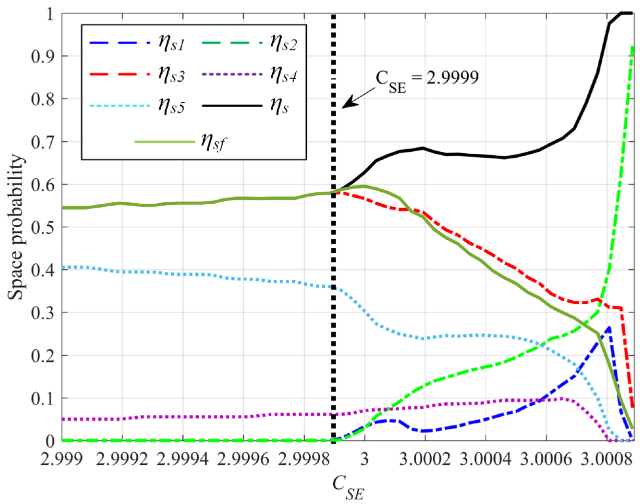

Figure 15. Curves of space probabilities with different C_{SE}.

Combining the space and time conditions of STC, we propose a STC index for a given C_{SE} as follows:

$$\lambda_{\mathrm{stc}}(C_{\mathrm{SE}}) = \frac{\iint_{U_c} \eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1)\mathrm{d}\alpha_1 \mathrm{d}\beta_1}{m(\Omega)}. \quad (7)$$

λ_{stc} combines the time and space probabilities of STC into one index, and can help us comprehensively evaluate the influence of C_{SE} on STC. Unfortunately, the rigorous calculations of $\eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1)$ in U_c need massive numerical integrations in the PBCM. Comparing Figs 10 and 14, we observe that the values of η_t in most regions of U_1, U_2, U_4, and U_5 can be approximately regarded as four constants: 0.9, 0.8, 0.01, and 0, respectively. However, the value of η_t in most of the region of U_3, as mentioned in the previous section, decreases with the decrease of C_{SE}. To reduce the computation cost, the following approximate expression of $\eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1)$ is used. For a given C_{SE},

$$\eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1) = \begin{cases} 0.9, & \text{if} \quad (\alpha_1, \beta_1) \in U_1; \\ 0.8, & \text{if} \quad (\alpha_1, \beta_1) \in U_2; \\ h(C_{\mathrm{SE}}), & \text{if} \quad (\alpha_1, \beta_1) \in U_3; \\ 0.01, & \text{if} \quad (\alpha_1, \beta_1) \in U_4; \\ 0, & \text{if} \quad (\alpha_1, \beta_1) \in U_5; \end{cases} \quad (8)$$

where

$$h(C_{\mathrm{SE}}) = \left(15^{2255.639(C_{\mathrm{SE}}-2.9995)} + 27.353\right)/5670.588.$$

The derivation of the expression $h(C_{\mathrm{SE}})$ in U_3 is given in Appendix B. Fig. 16 shows the curves of $\bar{\eta}_t$ and $\lambda_{\mathrm{stc}}(C_{\mathrm{SE}})$ with different C_{SE} using equation (8). The circle and star points denote numerical results in the PBCM. As we can see from the figure, when C_{SE} is smaller than 2.9999, $\bar{\eta}_t$ and λ_{stc} are both quite small; when C_{SE} is larger than 2.9999, $\bar{\eta}_t$ significantly increases with the increase of C_{SE}, but the change of λ_{stc} is relatively slow. This phenomenon can be explained by Fig. 10. When C_{SE} is smaller than 2.9999, only U_3 with small η_t is suitable for STC, so both $\bar{\eta}_t$ and λ_{stc} are near 0. When C_{SE} is larger than 2.9999, with the increase of C_{SE}, the space probabilities of U_1 and U_2 with the higher η_t gradually increase, so $\bar{\eta}_t$ increases significantly. However, as we can see from Fig. 10, U_c will shrink when C_{SE} increases, so the increase of $\bar{\eta}_t$ is offset by the reduction of U_c, and λ_{stc} maintains in a stable stage. Based on the curve of λ_{stc}, the most suitable and possible range of C_{SE} for STC is

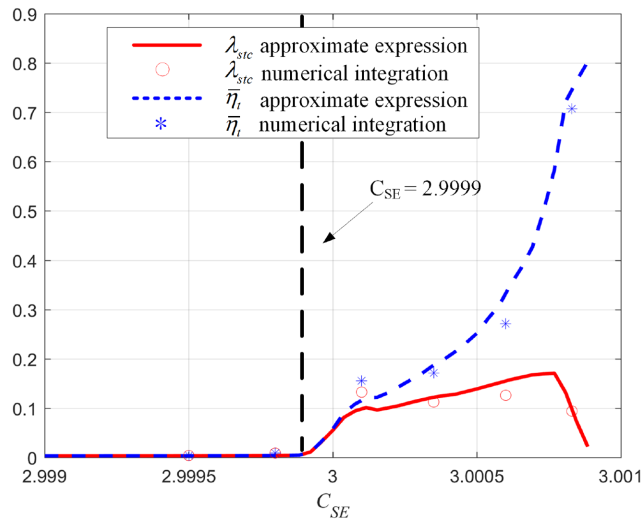

Figure 16. Curves of $\bar{\eta}_t$ and $\lambda_{\mathrm{stc}}(C_{\mathrm{SE}})$ with different C_{SE}.

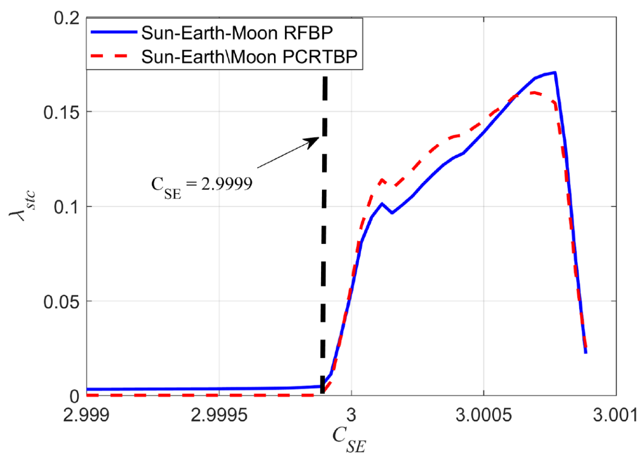

Figure 17. Curves of λ_{stc} in the Sun–Earth/Moon PCRTBP and the Sun–Earth–Moon PBCM.

[2.9999, 3.000 89], which is consistent with the result of the TCO (Granvik et al. 2012). However, only few NEAs have C_{SE} in this range, including asteroid 2006 RH_{120} (Sánchez & Yárnoz 2016).

The influence of the Moon on STC can be studied by the STC indices λ_{stc} in the Sun–Earth/Moon PCRTBP and the Sun–Earth–Moon RFBP. In the former model, the Earth and the Moon are combined into a single primary located at the EMB with the combined mass of the Earth and the Moon, so the influence of the Moon on STC does not exist. Based on the analysis in Section 3, $\eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1)$ in the Sun–Earth/Moon PCRTBP can be expressed as

$$\eta_t(C_{\mathrm{SE}}, \alpha_1, \beta_1) = \begin{cases} 1, & if \quad (\alpha_1, \beta_1) \in U_1 \cup U_2; \\ 0, & if \quad (\alpha_1, \beta_1) \in U_3 \cup U_4 \cup U_5. \end{cases} \quad (9)$$

Using equations (7) and (9), λ_{stc} in the Sun–Earth/Moon PCRTBP can be computed approximately (see Fig. 17). As we can see, when C_{SE} is larger than 2.9999, λ_{stc} in the Sun–Earth/Moon PCRTBP has no significant difference from that in the Sun–Earth–Moon PRFBP, so the presence of the Moon does not efficiently improve the likelihood of STC. However, when C_{SE} is smaller than 2.9999, λ_{stc} of the Sun–Earth/Moon PCRTBP decreases to 0, but λ_{stc} of the Sun–Earth–Moon PRFBP is still larger than 0. Therefore, we conclude that a LGA can increase the probability of STC for a

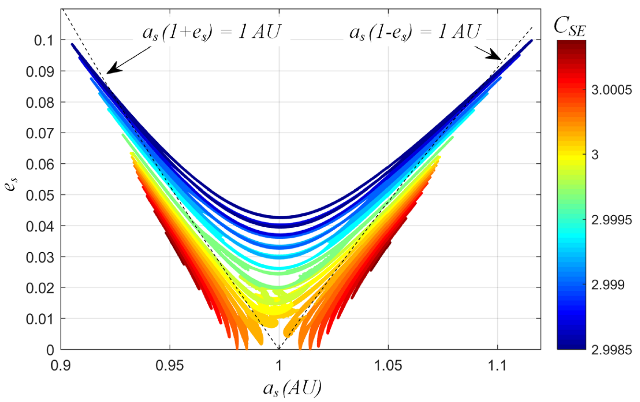

Figure 18. Distribution of a_s and e_s of the STC orbits for different C_{SE}.

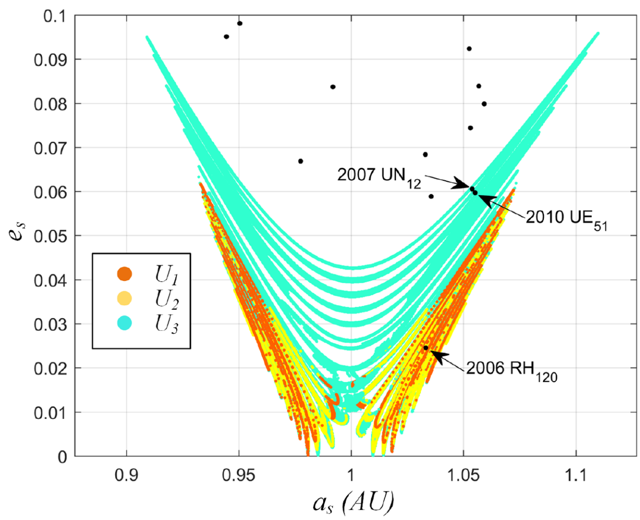

Figure 19. Potential asteroids in (a_s, e_s) coordinates.

smaller C_{SE} or faster objects, which is in accord with the result of Granvik et al. (2012). Since most NEAs have smaller C_{SE} (Sánchez & Yárnoz 2016), the LGA is very important in the endgame of the asteroid capture mission.

6 POTENTIAL STC ASTEROIDS

In this section, potential asteroids satisfying the requirements of STC are discussed. As mentioned in the previous section, the subsets $\bigcup_{i=1}^{3} U_i$ are regarded as suitable regions for STC in U_c. Using the points in these subsets as the initial points of backward integration, we can obtain the states of the corresponding STC orbits on the boundary of the EMS. Then, the state in the Sun–EMB rotating frame can be transformed into orbital elements in the heliocentric inertial frame (Chobotov 2002). It should be noted that the numerical integration is undertaken in the Sun–Earth/Moon PCRTBP rather than the PBCM, because the former can approximately replace the latter outside the SOIEM (Qi & Xu 2015).

Table 2. Orbital elements of the potential asteroids.

Asteroid	M (deg)	θ_e (deg)	θ_m (deg)
2010 UE_{51}	56.135 14	0	39.294
2007 UN_{12}	50.365 06	0	105.400
(2006 RH_{120})$_1$	91.426 45	0	223.725
(2006 RH_{120})$_2$	91.426 45	0	73.825

Fig. 18 displays the distribution of the semimajor axis a_s and the eccentricity e_s of the STC orbits on the boundary of the EMS, where the colours denote the magnitudes of C_{SE}. The two dashed lines in the figure are the lines of $a_s(1 + e_s) = 1$ au and $a_s(1 - e_s) = 1$ au. As mentioned in Section 5, the mean time probability of STC $\bar{\eta}_t$ decreases with the decrease of C_{SE}, so we find that the points (a_s, e_s) with high probability of STC are mostly located below the dashed lines.

Using Fig. 18, we can choose the potential STC asteroids from the asteroid data base. The data (https://www.minorplanetcenter.net/data, Retrieved 2017 December 12) of the NEAs can be obtained from the Minor Planet Center of the International Astronomical Union. Granvik et al. (2012) pointed out that the likelihood of a close Earth encounter is highest for Earth-like orbits with $a_s \sim 1$ au, $e_s \sim 0$, and $i_s \sim 0$, so 13 NEAs with 0.9 au $< a_s <$ 1.1 au, $e_s < 0.1$, and $i_s < 1^\circ$ are selected from 17083 NEAs, and their (a_s, e_s)s are plotted in Fig. 19. As we can see from the figure, there are three potential STC asteroids located in the STC capture region. The orbital elements of three potential STC asteroids are listed in Table 1. Since asteroid 2006 RH_{120} is located in the U_2 region, we predict that it will be temporarily captured again by the EMS in the future, and its STC probability is quite high. Asteroids 2007 UN_{12} and 2010 UE_{51} are located in the U_3 region, so only via a LGA can they be temporarily captured by the EMS, and their STC probabilities are small.

Since the inclinations of the three potential STC asteroids are smaller than 1° (see Table 1), we assume that the orbital planes of the Sun, Earth, Moon, and asteroid are same. Then, using the orbital elements in Table 1, the STC capture orbits of the asteroids can be obtained by numerical integration in the PBCM.

When the angles of the asteroid, Earth, and Moon in the heliocentric inertial frame at the initial moment of the numerical integration are set as the values in Table 2, the corresponding STC orbits can be obtained in the PBCM (see Fig. 20). In Table 2, θ_e denotes the angle between the Earth and the x-axis of the J2000.0 heliocentric frame; θ_m denotes the phase angle of the Moon in the Sun–EMB rotating frame, similar to θ_{m0}; M is the mean anomaly of the asteroid at the initial moment. Since (a_s, e_s) of asteroid 2006 RH_{120} is located in U_2, it has two methods to realize a STC: by the dynamic system of the Sun–Earth/Moon CRTBP directly and via a LGA. Two kinds of STC orbits of asteroid 2006 RH_{120} are shown in Fig. 20 and denoted

Table 1. Orbital elements of the potential asteroids.

Epoch: JD 2458000.5 (2017.9.4)						
Asteroid	a_s (au)	e_s	i_s (deg)	ω (deg)	Ω (deg)	M (deg)
2010 UE_{51}	1.055 1554	0.059 6890	0.624 10	47.199 91	32.303 81	76.135 14
2007 UN_{12}	1.053 8197	0.060 4933	0.235 53	134.484 53	216.008 65	78.265 06
2006 RH_{120}	1.033 1489	0.024 4877	0.594 97	10.032 57	51.175 01	110.626 45

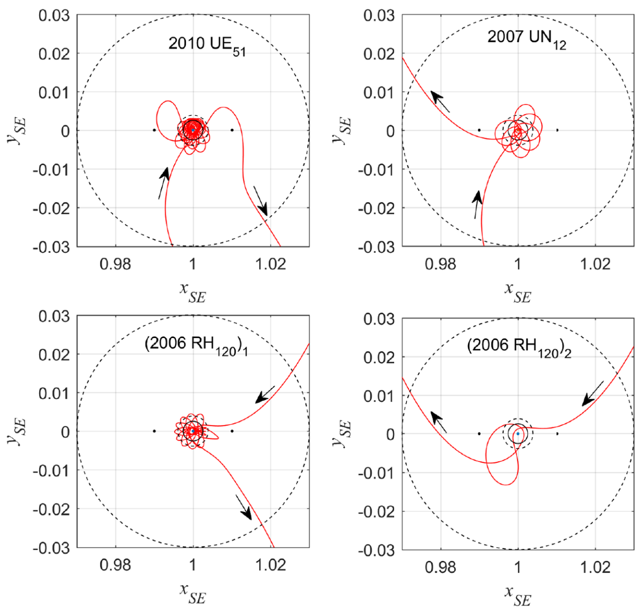

Figure 20. Four STC orbits of the potential asteroids in the Sun–EMB rotating frame.

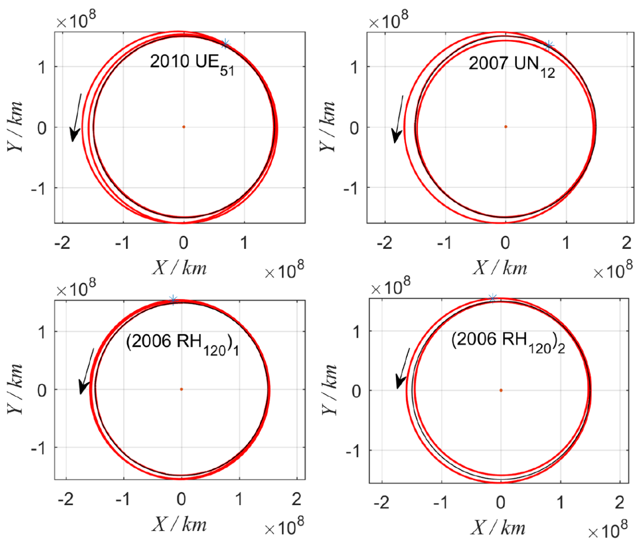

Figure 21. Four STC orbits of the potential asteroids in the heliocentric inertial frame.

by $(2006\ \mathrm{RH}_{120})_1$ and $(2006\ \mathrm{RH}_{120})_2$, respectively. They have the same M but different θ_m (see Table 2). The orbits in Fig. 20 are described in the Sun–EMB rotating frame. The outer dashed circle is the boundary of the EMS, i.e. three Hill-radius distance. The inner dashed and solid circles denote the SOIEM and the Moon orbit, respectively. As we can see from Fig. 20, $(2006\ \mathrm{RH}_{120})_1$ uses a LGA to achieve STC, whereas $(2006\ \mathrm{RH}_{120})_2$ can realize a STC directly without a lunar flyby. As for the orbits of 2007 UN_{12} and 2010 UE_{51}, as mentioned before, only via a LGA can they be temporarily captured by the EMS (see Fig. 20).

The four STC orbits of Fig. 20 are illustrated in Fig. 21 in the heliocentric inertial frame, where the star points denote the initial points of the numerical integration and the black circles are the EMB

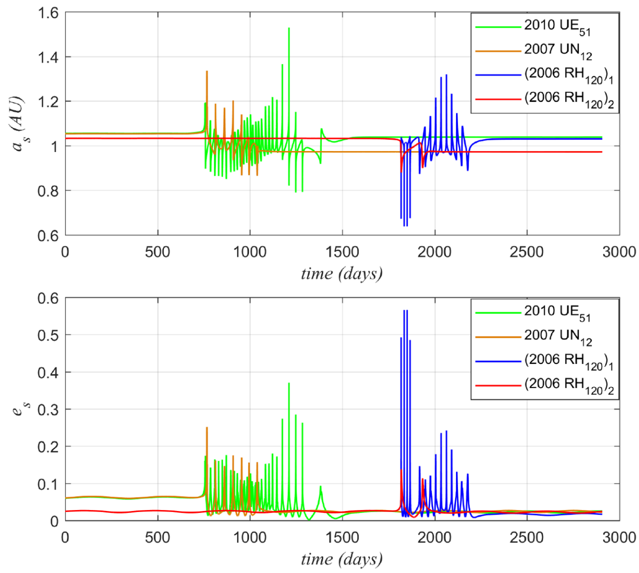

Figure 22. Curves of a_s and e_s of four STC orbits.

orbit around the Sun. As we can see from the figure, the shapes of the asteroids' orbits are changed after the STC phases. The more obvious evidence is provided by the curves of a_s and e_s of the four STC orbits in Fig. 22. From the figures, we find that the curves of a_s and e_s fluctuate dramatically during the STC phases, especially when the lunar flyby occurs. The orbital elements of orbits 2007 UN_{12} and $(2006\ \mathrm{RH}_{120})_2$ have significant differences before and after STC. In addition, the numerical calculation indicates that the duration of the four STC orbits, 2010 UE_{51}, 2007 UN_{12}, $(2006\ \mathrm{RH}_{120})_1$, and $(2006\ \mathrm{RH}_{120})_2$, in the region of the EMS are 834.8, 359.9, 517, and 241.5 d, respectively. Therefore, we have adequate time or chance to perform manoeuvres to achieve long-term capture in the EMS. The design of low-energy transfers from STC to long-term capture will be our future work.

Using the state of $(2006\ \mathrm{RH}_{120})_1$ on the boundary of the EMS as the initial state, we examine the STC orbits in the ephemeris model. Fig. 23 shows two STC orbits in the Sun–EMB dimensionless rotating frame with the same initial state but different initial moments, where the red trajectory starts at 2019 MAR 27 22:20:12.425740 (UTC), and the blue dashed trajectory starts at 2019 APR 06 05:45:28.758491 (UTC). The length unit of the Sun–EMB rotating frame is the distance between the Sun and the EMB, so its value is not constant in the ephemeris model. The green line in this figure represents the Moon's orbit. Due to the inclination of the Moon, the Moon's orbit is spatial in the Sun–EMB rotating frame (see the green line in Fig. 23b). The red trajectory achieves STC after the first lunar flyby, and is bounded in the EMS for more than 600 d. Since the position of the Moon is outside the x_{SE}–y_{SE} plane during the first lunar flyby, the red STC orbit is also out of the plane. However, the blue dashed trajectory can directly realize STC without any lunar flyby, so its inclination is quite small.

7 CONCLUSIONS

In this paper, the STC of asteroids in the EMS was proposed and studied. Using the Poincare section of the invariant manifolds of the LPOs on the SOIEM, the space condition of STC was preliminarily analysed in the Sun–Earth/Moon PCRTBP. For a given Jacobi constant C_{SE}, the five subsets, U_1–U_5, of the feasible region

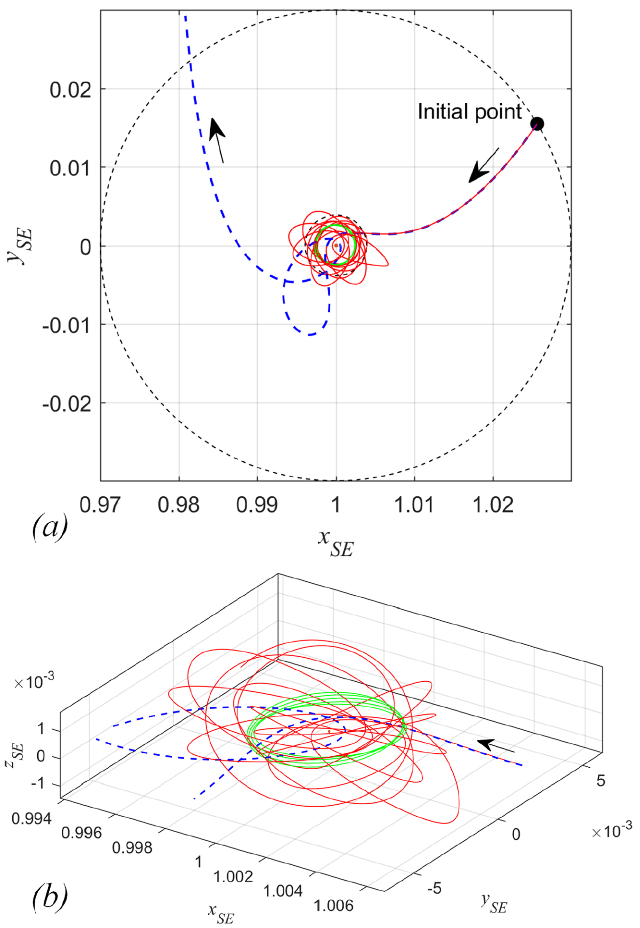

Figure 23. Two STC orbits in the Sun–EMB dimensionless rotating frame with the same initial state but different initial moments: (a) view from the north ecliptic pole, (b) detail in the region near the Moon's orbit.

U_c were defined and discussed. Then, the time condition of STC was studied by scanning the parameter in the PBCM. For a given $C_{\rm SE}$, the distributions of the STC windows of U_1–U_5 were analysed and compared. The time probability of STC was defined as the ratio of the length of the STC windows to the length of the full range. Numerical results indicated that the time probability of STC gradually decreases from U_1 to U_5. In this way, the space condition of a STC in an autonomous system is connected with the time condition of a STC in a time-varying system. Then, the influence of $C_{\rm SE}$ on STC was discussed by the space and time probabilities of STC. Numerical calculation demonstrated that the absolute space probability of STC decreases when $C_{\rm SE}$ is larger than 3 due to the reduction of U_c, but the mean time probability of STC with $C_{\rm SE}$ larger than 2.9999 is significantly larger than that with $C_{\rm SE}$ smaller than 2.9999. Combining the space and time probabilities of STC, we proposed a STC index to comprehensively evaluate probability of STC. Numerical results indicated that the most suitable and possible range of $C_{\rm SE}$ for STC is [2.9999, 3.000 89], which is consistent with the result of the TCO (Granvik et al. 2012). In addition, based on the analysis of the STC index, we found that the presence of the Moon does not efficiently improve the likelihood of STC for large $C_{\rm SE}$, but the LGA can increase the probability of STC for smaller $C_{\rm SE}$ or faster objects. Since most NEAs have smaller $C_{\rm SE}$, the LGA is very important in the endgame of the asteroid capture mission. Finally, three potential STC asteroids were found and analysed.

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APPENDIX A: DERIVATION OF THE ANGLES IN THE SOIEM

Given $C_{\rm SE}$, α_1, and β_1, the position coordinates of the asteroid entering the SOIEM in the Sun–EMB dimensionless rotating frame can be obtained by

$$x = r_{\rm soiem}\cos\alpha_1/L_{\rm se} + 1 - \mu_{\rm se},$$
$$y = r_{\rm soiem}\sin\alpha_1/L_{\rm se}, \tag{A1}$$

where L_{se} is the distance between the Sun and the EMB, approximately $1.494\,61 \times 10^8$ km, and μ_{se} is the mass ratio of the Sun–Earth/Moon CRTBP, $3.040\,424 \times 10^{-6}$. The magnitude of velocity in the Sun–EMB rotating frame can be expressed as

$$v = \sqrt{(x^2 + y^2) + 2U - C_{\mathrm{SE}}}, \tag{A2}$$

where U is the effective potential,

$$U = \frac{1-\mu_{se}}{r_1} + \frac{\mu_{se}}{r_2} + \frac{1}{2}\mu_{se}(1-\mu_{se}),$$
$$r_1 = \sqrt{(x+\mu_{se})^2 + y^2},$$
$$r_2 = \sqrt{(x+\mu_{se}-1)^2 + y^2}.$$

The velocity coordinates in the Sun–EMB rotating frame can be obtained by

$$\dot{x} = -v\sin(\alpha_1 - \beta_1),$$
$$\dot{y} = v\cos(\alpha_1 - \beta_1). \tag{A3}$$

Furthermore, the state in the EMB-centric rotating frame can be easily obtained,

$$x_{e1} = x - 1 + \mu_{se},$$
$$y_{e1} = y,$$
$$\dot{x}_{e1} = \dot{x},$$
$$\dot{y}_{e1} = \dot{y}. \tag{A4}$$

The transformation matrix from EMB-centric rotating frame to the EMB-centric inertial frame can be expressed as

$$T(\varphi)=\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \quad \dot{T}(\varphi)=\begin{pmatrix} -\sin\varphi & -\cos\varphi \\ \cos\varphi & -\sin\varphi \end{pmatrix}. \tag{A5}$$

We assume that when the asteroid first enters the SOIEM, the x-axes of the inertial frame and the rotating frame coincide, i.e. $\varphi = 0$. Then, the state in the EMB-centric inertial frame can be obtained by

$$\begin{pmatrix} X \\ Y \end{pmatrix} = T(0)\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix},$$
$$\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix} = \dot{T}(0)\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix} + T(0)\begin{pmatrix} \dot{x}_{e1} \\ \dot{y}_{e1} \end{pmatrix}. \tag{A6}$$

We assume that the orbit in the SOIEM can be approximated by a conic orbit in the EMB-centric two-body model. Then, based on the classical formulas of the two-body model (Chobotov 2002), the semimajor axis a, the eccentricity e, and the distance of the periapsis and the EMB r_p can be calculated from the state in the inertial frame. The orbital coordinates of the conic orbit are defined, where the x-axis is the vector from the EMB to the periapsis of the conic (see Fig. A1). The angle between two x-axes is denoted by γ, and $\gamma = \alpha_1 \pm f_{\mathrm{soiem}}$, where f_{soiem} is the true anomaly of the entering point on the SOIEM, and the sign $\pm$ depends on the direction of the orbit: clockwise motion corresponds to +; anticlockwise motion corresponds to −. The state at the entering point in the orbital coordinates can be obtained by

$$\begin{pmatrix} \bar{X} \\ \bar{Y} \end{pmatrix} = T^{-1}(\gamma)\begin{pmatrix} X \\ Y \end{pmatrix}, \quad \begin{pmatrix} \dot{\bar{X}} \\ \dot{\bar{Y}} \end{pmatrix} = T^{-1}(\gamma)\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix}. \tag{A7}$$

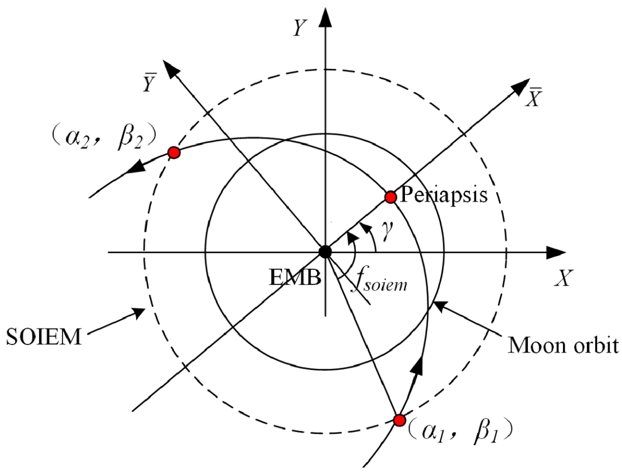

Figure A1. EMB-centric inertial coordinates and orbital coordinates.

The state of the outgoing point of the orbit on the SOIEM can be derived as

$$\begin{pmatrix} \bar{X}_1 \\ \bar{Y}_1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \bar{X} \\ \bar{Y} \end{pmatrix} = PT^{-1}(\gamma)\begin{pmatrix} X \\ Y \end{pmatrix},$$
$$\begin{pmatrix} \dot{\bar{X}}_1 \\ \dot{\bar{Y}}_1 \end{pmatrix} = -\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} \dot{\bar{X}} \\ \dot{\bar{Y}} \end{pmatrix} = -PT^{-1}(\gamma)\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix}, \tag{A8}$$

where $P = [10; 0 - 1]$. According to equations (A7) and (A8), the state at the outgoing point (α_2, β_2) in the EMB-centric inertial frame can be obtained as

$$\begin{pmatrix} X_1 \\ Y_1 \end{pmatrix} = T(\gamma)PT^{-1}(\gamma)\begin{pmatrix} X \\ Y \end{pmatrix},$$
$$\begin{pmatrix} \dot{X}_1 \\ \dot{Y}_1 \end{pmatrix} = -T(\gamma)PT^{-1}(\gamma)\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix}. \tag{A9}$$

Then, based on equations (A6) and (A9), the state of the outgoing point of the orbit in the EMB-centric rotating frame can be expressed as

$$\begin{pmatrix} x_{e2} \\ y_{e2} \end{pmatrix} = T^{-1}(\delta)T(\gamma)PT^{-1}(\gamma)T(0)\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix} = S\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix},$$

$$\begin{pmatrix} \dot{x}_{e2} \\ \dot{y}_{e2} \end{pmatrix} = -T^{-1}(\delta)\begin{pmatrix} \dot{X}_1 \\ \dot{Y}_1 \end{pmatrix} - T^{-1}(\delta)\dot{T}(\delta)\begin{pmatrix} x_{e1} \\ y_{e1} \end{pmatrix} = -S\begin{pmatrix} \dot{x}_{e1} \\ \dot{y}_{e1} \end{pmatrix}, \tag{A10}$$

where

$$S = \begin{pmatrix} \cos(2\gamma-\delta) & \sin(2\gamma-\delta) \\ \sin(2\gamma-\delta) & -\cos(2\gamma-\delta) \end{pmatrix}$$

and δ is the rotation angle of the EMB-centric rotating frame in the EMB-centric inertial frame during the flight time in the SOIEM. Therefore, we can get $\delta = 4\pi\Delta t_{\mathrm{soiem}}/T_{\mathrm{se}}$, where $\Delta t_{\mathrm{soiem}}$ and T_{se} denote the flight time from the entering point to the periapsis and the period of the EMB around the Sun in the Sun–EMB rotating frame, respectively. $\Delta t_{\mathrm{soiem}}$ can be derived from the orbital elements of the conic orbit (Chobotov 2002). Using equations (A1), (A3), (A4), and (A10), we can derive

$$\alpha_2 = \alpha_1 \pm 2 f_{\mathrm{soiem}} - \delta,$$
$$\beta_2 = -\beta_1. \tag{A11}$$

In the same way, we can also get

$$\theta_{c1} = \alpha_1 \pm (f_{\mathrm{soiem}} - f_{\mathrm{moon}}) - 2\pi(\Delta t_{\mathrm{soiem}} - \Delta t_{\mathrm{moon}})/T_{\mathrm{se}},$$
$$\theta_{c2} = \alpha_1 \pm (f_{\mathrm{soiem}} + f_{\mathrm{moon}}) - 2\pi(\Delta t_{\mathrm{soiem}} + \Delta t_{\mathrm{moon}})/T_{\mathrm{se}},$$
$$\theta_{m1} = \theta_{c1} - \omega_m(\Delta t_{\mathrm{soiem}} - \Delta t_{\mathrm{moon}}),$$
$$\theta_{m2} = \theta_{c2} - \omega_m(\Delta t_{\mathrm{soiem}} + \Delta t_{\mathrm{moon}}), \tag{A12}$$

where f_{moon} and Δt_{moon} denote the magnitude of the true anomaly of the first crossing point of the conic orbit at the Moon orbit and the flight time from the first crossing point at the Moon orbit to the periapsis, respectively, both which can be obtained from the orbital elements of the conic orbit (Chobotov 2002). ω_m is the angular velocity of the Moon in the Sun–EMB rotating frame, 0.212781 rad d^{-1}. The determination of $\pm$ in equation (A12) is similar the discussion about γ.

APPENDIX B: APPROXIMATE EXPRESSION IN U_3

Based on the numerical results in Fig. 14, the mean values of η_t in U_3 are estimated and displayed in Fig. B1. From the figure, we infer that the approximate expression of η_t in U_3 should be a function with some exponential type. Numerical test indicates that the shape of the function $y = 15^x$ in the range $[0, 3]$ is similar to the distribution of the estimated values, so the approximate expression of $\eta_t(C_{\mathrm{SE}})$ in U_3 can be expressed as

$$y = k_1\eta_t + b_1 = 15^x = 15^{k_2(C_{\mathrm{SE}}-2.9995)+b_2}. \tag{B1}$$

From Fig. B1, two points where $\eta_t(2.9995) = 0.005$ and $\eta_t(3.000\,83) = 0.6$ are chosen as the fitting points. Then, we can get $k_1 = 5670.588$, $k_2 = 2255.639$, $b_1 = -27.353$, and $b_2 = 0$. Therefore, the approximate expression of η_t in U_3 can be expressed as

$$h(C_{\mathrm{SE}}) = \eta_t(C_{\mathrm{SE}}, U_3) = \left(15^{2255.639(C_{\mathrm{SE}}-2.9995)} + 27.353\right)/5670.588. \tag{B2}$$

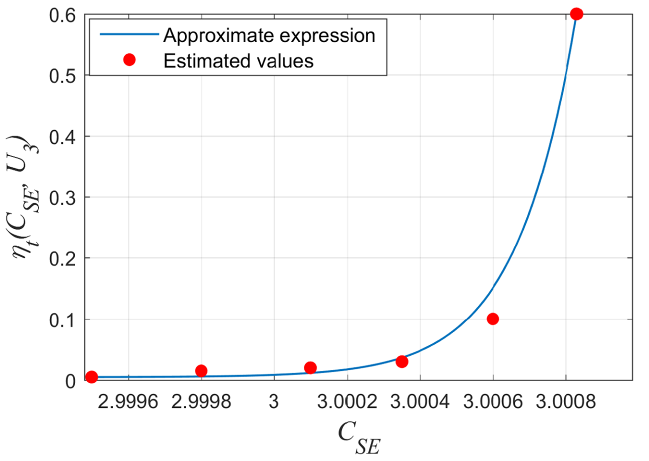

Figure B1. Approximate curve of η_t and estimated values in U_3.

As we can see from Fig. B1, the fitting curve based on equation (B2) well approximates the estimated values.

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